Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 22: Approved Solutions

Direct Proof

Covers Chartrand & Zhang, Section 3.2

Status of this set

Ungraded practice. Solutions released with it. Write every proof in complete sentences — that is the whole exercise.

Part I: How to Practice This Material

Four steps, every time

The template

1. **Assume the hypothesis** and write it as an equation using the definition. "Let n be odd. Then $n = 2k + 1$ for some integer k."
2. **Substitute** into the expression you must analyse.
3. **Algebra** until the target form appears.
4. **Factor out the 2**, say the remaining factor is an integer, and conclude.

Step 4 is where the proof finishes. Everything before is arithmetic.

The definitions, exactly

n is **even** if $n = 2a$ for some *integer* a.
n is **odd** if $n = 2b + 1$ for some *integer* b.
You may assume that sums, products, and negatives of integers are integers. You may *not* assume "even plus even is even" — that is a theorem, not an axiom.

Three mistakes to expect

1. **Omitting "for some integer k."** The last line depends on it.
2. **Stopping at $2m - 1$.** Odd means $2b + 1$. Adjust the algebra.
3. **Using the same letter for two independent quantities.** If $m = 2a$ and $n = 2a$, you have assumed $m = n$.

Part II: Practice Problems

Section A — Parity

1. Prove: if n is an odd integer, then $7n + 4$ is odd.

Approved solution

Proof. Let n be an odd integer. Then $n = 2k + 1$ for some integer k. Therefore

$$7n + 4 = 7(2k + 1) + 4 = 14k + 7 + 4 = 14k + 11 = 2(7k + 5) + 1.$$

Since $7k + 5$ is an integer, $7n + 4$ is odd. ■

Watch step 4. $14k + 11$ must be rewritten as $2(\cdot) + 1$, so peel off the 1: $14k + 11 = 14k + 10 + 1$. Writing $2(7k + 5.5)$ would be fatal — $7k + 5.5$ is not an integer.

2. Prove: if n is an even integer, then $3n^2 + n$ is even.

Approved solution

Proof. Let n be even, so $n = 2k$ for some integer k. Then

$$3n^2 + n = 3(2k)^2 + 2k = 12k^2 + 2k = 2(6k^2 + k).$$

Since $6k^2 + k$ is an integer, $3n^2 + n$ is even. ■

Note that both terms already carried a factor of 2, so no adjustment was needed. That is not always the case — compare the next problem.

3. Prove: if n is an odd integer, then $n^2 + 3$ is even.

Approved solution

Proof. Let n be odd, so $n = 2k + 1$ for some integer k. Then

$$n^2 + 3 = (2k + 1)^2 + 3 = 4k^2 + 4k + 1 + 3 = 4k^2 + 4k + 4 = 2(2k^2 + 2k + 2).$$

Since $2k^2 + 2k + 2$ is an integer, $n^2 + 3$ is even. ■

The $+1$ from expanding the square combined with the $+3$ to give 4, which is even. Watching constants combine is most of the work in these.

Section B — Two variables

4. Prove: if m and n are both odd integers, then $m + n$ is even.

Approved solution

Proof. Let m and n be odd. Then $m = 2a + 1$ and $n = 2b + 1$ for **some integers** a **and** b. Therefore

$$m + n = (2a + 1) + (2b + 1) = 2a + 2b + 2 = 2(a + b + 1).$$

Since $a + b + 1$ is an integer, $m + n$ is even. ■

The two letters matter. Writing $m = 2a + 1$ and $n = 2a + 1$ would assert $m = n$, which the claim does not say. Independent quantities need independent letters — this is the third common error from Part I, and it appears the moment a second variable does.

5. Prove: if m is even and n is odd, then mn is even.

Approved solution

Proof. Let m be even and n be odd. Then $m = 2a$ and $n = 2b+1$ for some integers a and b. Therefore

$$mn = 2a(2b + 1) = 2\big(a(2b + 1)\big).$$

Since $a(2b + 1)$ is an integer, mn is even. ■

Here n never had to be unfolded at all — the factor of 2 from m carries the whole argument. **Unfold only what you need.** A shorter proof is a better proof.

Section C — Beyond parity

6. Prove: if r and s are rational numbers, then rs is rational.

Approved solution

Proof. Let r and s be rational. Then $r = a/b$ and $s = c/d$ where $a, b, c, d \in \mathbb{Z}$ with $b \neq 0$ and $d \neq 0$. Therefore

$$rs = \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

Since ac and bd are integers, and $bd \neq 0$ because a product of nonzero numbers is nonzero, rs is rational. ■

Do not skip the $bd \neq 0$ step. The definition of rational requires a nonzero denominator, so the proof must supply one.

Section D — Writing

7. In five to eight complete sentences, explain why a direct proof of $P \Rightarrow Q$ only needs to consider the case where P is true. Refer to the truth table for the implication.

Approved solution

Model answer. An implication $P \Rightarrow Q$ is false in exactly one row of its truth table: the row where P is true and Q is false. In the two rows where P is false the implication is true automatically, regardless of Q, so nothing can go wrong there and nothing needs to be checked. That leaves only the case where P holds, and in that case the implication is true precisely when Q is also true. So proving the statement reduces to showing that Q follows whenever P is assumed, which is exactly what a direct proof does. The opening line "assume P" is therefore not a convenience but a complete reduction of the problem: the rows it ignores were never at risk.

Full credit requires identifying the single false row and explaining why the others need no attention. An answer that only describes the template earns about half.

Section E — Challenge (optional)

8. Prove: if n is an odd integer, then n^3 is odd. Then say what your proof shows about n^4 when n is odd, without a new computation.

Approved solution

Proof. Let n be odd, so $n = 2k + 1$ for some integer k. Then

$$n^3 = (2k+1)^3 = 8k^3 + 12k^2 + 6k + 1 = 2(4k^3 + 6k^2 + 3k) + 1.$$

Since $4k^3 + 6k^2 + 3k$ is an integer, n^3 is odd. ■

About n^4. We now know two things: n odd implies n^2 odd (Lecture 23), and a product of two odd integers is odd. Since $n^4 = n^2 \cdot n^2$ and n^2 is odd, n^4 is odd — no expansion required.

This is what "once proved, may be used" means. The first parity results are earned with algebra; after that, they become tools, and later proofs get shorter. Expanding $(2k+1)^4$ would work and would be a waste of effort.

Instructor notes

- **Problem 4 is the diagnostic.** The moment a second variable appears, some students reuse the same letter and unknowingly assume $m = n$. Catch it here, before Homework 9.
- **Problem 5** rewards restraint: n never needs unfolding. Students who expand both get the right answer the long way.
- **Problem 8's second half** is the first time students reuse an earlier result instead of recomputing. That habit is what makes Chapter 4 feasible.
- **Problem 6's** $bd \neq 0$ is routinely skipped. It is part of the definition.
- Reserved for Homework 9: Section 3.2 even-numbered exercises. Odd ones have printed answers and are self-check only.